

Mars

Overview

Mars is the fourth planet from the Sun, orbiting at an average distance of 228 million km and completing one orbit every 687 Earth days. With a diameter of 6,779 km — about half that of Earth — Mars is the second smallest planet in the Solar System after Mercury. Its thin atmosphere, composed of 95 % carbon dioxide, 2.6 % nitrogen, and 1.9 % argon, provides little insulation or protection from radiation. Surface pressure on Mars is less than 1 % of Earth's, making it lethal to unprotected humans and too thin to maintain liquid water on the surface today. The planet's reddish hue is caused by iron oxide — rust — mixed into the fine dust that blankets much of the surface and is periodically lofted into planet-wide dust storms lasting for months.

Surface Features

Mars possesses the most dramatic topography in the Solar System. Olympus Mons is the tallest volcano and the tallest mountain in the Solar System, rising 21.9 km above the surrounding plains and spanning approximately 600 km in diameter — roughly the size of France. Its gentle slope means an observer standing at the base would not be able to see the summit, as the curvature of the planet would hide it below the horizon. The Valles Marineris is a system of canyons stretching over 4,000 km in length, up to 200 km wide and 7 km deep — long enough to span the entire continental United States and four times as deep as the Grand Canyon. Both poles of Mars are capped by layers of water ice and dry ice (frozen CO₂); the north polar cap is primarily water ice up to 3 km thick and about 1,000 km across. The southern hemisphere is older and more heavily cratered than the northern hemisphere, which is dominated by younger, smoother volcanic plains in a geological dichotomy whose origin remains one of the biggest unsolved mysteries in planetary science.

Water on Mars

Substantial evidence indicates that Mars had abundant liquid water on its surface billions of years ago, when its atmosphere was thicker and the climate warmer. Ancient dry riverbeds, valley networks, and lake delta formations — like the Jezero Crater delta currently being explored by the Perseverance rover — indicate that water once flowed and pooled on the Martian surface. Subsurface radar data from the MARSIS instrument aboard Mars Express detected a 20 km-wide zone of high radar reflectivity under the south polar ice cap in 2018, interpreted by some scientists as a subglacial lake of liquid briny water. Recurring Slope Lineae (RSL) — dark streaks that appear seasonally on steep slopes — were once thought to be flowing briny water, but are now believed more likely to be dry granular flows. The Mars Climate Database and multiple orbital missions confirm the presence of water ice just centimetres below the surface at mid-latitudes, potentially accessible to future human explorers as a resource.

Moons: Phobos and Deimos

Mars has two small, irregularly shaped moons: Phobos and Deimos, discovered in 1877 by American astronomer Asaph Hall. Phobos is the larger of the two, measuring about $27 \times 22 \times 18$ km, and orbits Mars at just 9,376 km above the surface — closer to its planet than any other moon in the Solar System — completing one orbit in just 7 hours and 39 minutes, faster than Mars itself rotates. Because of tidal forces, Phobos is spiralling inward and is expected to either crash into Mars or break apart into a ring system within 50 million years. Deimos orbits farther out at 23,460 km and is much smaller, measuring about $15 \times 12 \times 10$ km. Both moons are thought to be captured C-type asteroids from the outer asteroid belt, though some models suggest they could be remnants of a larger body that was disrupted in orbit around early Mars.

Exploration

Mars is the most explored planet beyond Earth, with over 50 missions attempted by various space agencies since the 1960s, though only about half have succeeded. NASA's Perseverance rover, which landed in Jezero Crater in February 2021, is the most sophisticated Mars rover ever built, carrying instruments to search for biosignatures, analyse rock chemistry, produce oxygen from the Martian atmosphere, and cache rock samples for future return to Earth. Accompanying Perseverance is the Ingenuity helicopter, which in April 2021 became the first powered, controlled aircraft to fly on another planet. ESA's ExoMars Trace Gas Orbiter has been studying the Martian atmosphere since 2018, searching for methane and other trace gases that could indicate biological or geological activity. Plans for human missions to Mars are being developed by NASA and SpaceX, with proposals targeting the 2030s and 2040s, though enormous engineering and physiological challenges — including radiation exposure and journey times of 6–9 months — remain to be solved.